

INTRODUCTION TO SIEM:

- Security Information and Event Management ^{system} (SIEM) is a security solution used by SOC analysts in the security operations center - res.
- In a network, all the devices continuously generate logs and these devices are called "log sources". These log sources are divided into two categories:
 - i) Host-Centric log source: These are the logs coming from a single machine (host). Ex: laptop, Desktop, server etc.
What logs they generate? Login logs, File access logs, Process logs, Application logs
 - ii) Network-Centric log source: These are the logs coming from the network devices. Ex: Firewall, router, switch, IDS/IPS.
What logs they generate? Incoming/Outgoing traffic, IP address, Blocked connections etc.
- SIEM is a security solution which collects logs from different log sources, standardize their format into a consistent one, correlates logs and detect malicious activities using detection rules.

Features of SIEM:

- i) Centralization of logs: SIEM collects logs from different ^{log} sources with the help of agents ~~and~~ or API ~~and~~. Agents are installed on each system and they send the logs from there to the SIEM solution.

ii) Normalization of logs: Logs are normalized into one consistent format + because logs from windows and linux or any other device will be in different format. Breaking down of a log into several fields for ease of understanding is known as parsing.

iii) Correlation of logs: Individual logs are not very useful, but correlating logs from different sources can form a totally different incident which can be harmful ^{and they look safe}

iv) Real-Time Alerting: SIEM detects malicious activities based on the detection rules saved in SIEM solution.

v) Dashboard and Reporting: A dashboard is a screen ~~where~~ in SIEM where you can see all security activity clearly such as Number of alerts, Failed logins, Active users, Rules triggered, Top domain visited etc. SIEM comes with default dashboard but you can create your custom dashboard.

- In windows logs can be viewed through event manager view.

In linux OS logs are stored in different directories:

i) `var/log/httpd`: Contains HTTP request/Response and error logs

ii) `var/log/cron`: Events related to cron jobs

iii) `var/log/auth.log` and `/var/log/secure`: Stores authentication logs

iv) `var/log/kern`: It stores kernel logs

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In linux apache-related logs are `var/log/httpd` or `var/log/apache`

- Logs ingestion:

Each SIEM solution has its own way of ingesting logs. Some common methods used by them are:

- i) Agent/Forwarder: These SIEM solutions use a lightweight tool ~~can~~ called an agent or forward which is installed on all devices and they capture the all logs and send them to SIEM server.
- ii) Syslog: It is a protocol that is used to collect logs from different systems and send it to centralized destination.
- iii) Manual upload: Some SIEM solutions also allow users to manually upload data for quick analysis.
- iv) Port-Forwarding: SIEM solutions can also receive data by port forwarding. In these SIEM solutions are set to listen data on a port and the ~~an~~ devices send data to the listening port.